Dynamical Origins of Speed–Accuracy Trade-offs and Dysfunctional Decision Regimes in a Reduced Cortical Decision Circuit

Ali Taghizadeh
Department of Computer Science,
Amirkabir University of Technology (Tehran Polytechnic)

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Abstract

This project is a purely computational study of perceptual decision-making in the reduced two-variable circuit model of Wong and Wang (2006) (DOI: https://doi.org/10.1523/JNEUROSCI.3733-05.2006). Two selective excitatory populations with slow NMDA-mediated recurrent excitation compete through shared inhibition and integrate noisy evidence over hundreds of milliseconds. I ask whether the behavior of this circuit (accuracy, reaction time, and breakdown into dysfunctional regimes) can be predicted from the deterministic geometry of its phase plane, and how recurrent excitation relative to cross-inhibition and background noise reshape that geometry. All results come from simulation and dynamical-systems analysis; no data collection is required.

1 Background and Research Question

In random-dot motion tasks, neurons in area LIP ramp their firing rates in a way consistent with accumulation of evidence to a threshold (Roitman and Shadlen, 2002). Attractor-network models explain this ramping as a transition from a spontaneous state, through the vicinity of a saddle point, into one of two choice attractors (Wang, 2002; Wong and Wang, 2006). The reduced model can also be approximated by a one-dimensional diffusion equation (Roxin and Ledberg, 2008) and relates to drift-diffusion models (Bogacz et al., 2006). Lam et al. (2022) studied how altered excitation–inhibition (E/I) balance affects decisions in the *spiking* circuit of Wang (2002), perturbing NMDA receptor function at pyramidal cells versus interneurons; they report that disruption in either direction impairs performance (an inverted-U dependence) and that the effects map onto an extended drift-diffusion model with self-coupling. The present project is complementary: it uses the *reduced* two-variable model, varies the effective ratio of self-excitation to cross-inhibition together with the noise amplitude, and links a systematic regime map to the deterministic bifurcation structure.

Research question. How do the ratio of recurrent excitation to cross-inhibition (J_{11}/J_{12}) and the background noise amplitude σ shape the phase-plane geometry of the decision circuit, and to what extent do deterministic geometric quantities predict stochastic behavior (accuracy, reaction time, regime)?

Testable hypotheses.

- **H1.** Across the (J_{12}, σ) plane, mean decision time and error rate at fixed coherence are explained by the unstable eigenvalue λ_u of the saddle, the distance d from the start state to the separatrix, and σ. This is tested by regression of $\ln \bar{T}_D$ and error rate on these quantities, relative to a σ-only model.
- **H2.** The boundaries between behavioral regimes (normal, impulsive, indecisive) coincide with the deterministic bifurcation curves, shifted by noise in a predictable direction.

Scope of H1. Two facts from the seed paper (Section 2) limit where H1 can hold. At zero coherence the start state lies exactly on the stable manifold, so $d \equiv 0$ and only λ_u is informative; d becomes testable at $c' > 0$. In addition, the paper reports that the saddle's time constants matter less as stimulus strength or coherence grows, because trajectories then rarely pass near the saddle. H1 is therefore tested mainly at low coherence, and a weaker effect at high coherence is itself an expected result. Negative or partial results will be reported and interpreted.

Terminology note. The two-variable model has no explicit inhibitory population. J_{11}/J_{12} is an effective ratio of self-excitation to cross-inhibition, and it is referred to as such rather than as an "E/I ratio".

2 Analysis of the Seed Paper

Reference. Wong, K.-F. and Wang, X.-J. (2006). A recurrent network mechanism of time integration in perceptual decisions. *Journal of Neuroscience*, 26(4), 1314–1328. DOI: https://doi.org/10.1523/JNEUROSCI.3733-05.2006.

2.1 Problem and contribution

Reaction times in a two-alternative motion-discrimination task are hundreds of milliseconds long, and LIP firing rates ramp over that time. The longest biophysical time constant in the circuit is the 100 ms NMDA decay, so the paper asks how a network can integrate evidence over a much longer time without fine-tuning. It also asks whether NMDA receptors are necessary for this, whether integration requires persistent activity as in working memory, and how the circuit relates mathematically to the diffusion model. The contribution is a two-variable model, derived from the spiking decision network of Wang (2002), that can be analyzed completely in a phase plane and that reproduces the psychometric and chronometric functions of Roitman and Shadlen (2002).

2.2 Model reduction

The reduction proceeds in two stages. First, a mean-field treatment of the spiking network gives 11 dynamical variables: firing rates and NMDA/AMPA/GABA gating variables for two selective excitatory pools, one non-selective excitatory pool, and one inhibitory pool, with an input–output function fitted to the first-passage-time formula of a leaky integrate-and-fire neuron. Second, three approximations reduce this to two variables:

1. the non-selective population fires at a constant 2 Hz;
2. the inhibitory population's transfer function is linearized (over its 8–15 Hz working range), so the inhibition can be absorbed into an effective cross-inhibition $-J_{12}S_2$;

3. the NMDA gating variable is far slower than all other variables, so those variables sit at their steady state and only S_1 and S_2 evolve.

Noise is added back to the mean-field model as an Ornstein–Uhlenbeck current with a 2 ms (AMPA) time constant. The version I implement is the Appendix model without AMPA at the recurrent synapses.

2.3 Main findings

- **Phase-plane picture.** Without a stimulus the model has five steady states: a low symmetric spontaneous state, two asymmetric persistent states, and two saddles. A stimulus removes the spontaneous state and creates a symmetric saddle; its stable manifold is the boundary between the two choice basins. Trajectories first drift along the diagonal and then fall toward one attractor, which gives a biphasic time course.
- **Why errors are slower.** With nonzero coherence the basin of the favored attractor grows. An error trial has to cross the stable manifold, which pushes it close to the saddle, where dynamics are slow. Correct trials stay away from the saddle and are faster.
- **Integration time.** Near the saddle the state obeys $\delta\mathbf{S}(t) = a_1\mathbf{v}_1 e^{-t/\tau_{stable}} + a_2\mathbf{v}_2 e^{t/\tau_{unstable}}$, and the time to leave a neighborhood of size δ_0 is $t_{stay} \approx \tau_{unstable}\ln(1/\delta_0)$. Decision time grows in parallel with $\tau_{unstable}$ as stimulus strength falls.
- **Critical coherence.** Above about 70% coherence the less-favored attractor annihilates with the saddle, so only one choice attractor remains and the choice is always correct regardless of noise.
- **Role of NMDA.** Raising the AMPA fraction at recurrent synapses shortens decision times but worsens accuracy, and beyond 50:50 the network becomes oscillatory. With AMPA only (and oscillations artificially suppressed), decision times above about 200 ms could hardly be reached even with fine-tuning.
- **Speed–accuracy trade-off via recurrent strength.** A larger recurrent strength w_+ gives faster but less accurate decisions across all coherences. This is the dynamical origin of the trade-off in the title of this project.
- **Regimes.** In the (μ_0, w_+) plane the paper finds monostable, competition and bistable regions, and four modes of operation: I (no decision, no memory), II (decision without memory), III (decision and working memory; the standard parameters), IV (no competition).
- **Relation to the diffusion model.** A one-dimensional reduction along the unstable manifold is only qualitatively valid (at small μ_0, where $\tau_{unstable} \gg \tau_{stable}$), since the stable-manifold phase of about 200–300 ms cannot be neglected. At realistic stimulus strengths $\tau_{stable} > \tau_{unstable}$ and a one-dimensional description fails.

2.4 Quantitative reference values

These values are targets for replication; they are reported for the paper's standard parameter set, and the Appendix parameter set is expected to behave similarly but not identically.

Quantity	Value reported in the paper
Weibull threshold α, slope β	Reduced model 7.2%, 1.25; monkey 7.4%, 1.3; spiking network 8.4%, 1.6
Decision criterion	Threshold $\theta = 15$ Hz (corresponds to $S \approx 0.49$); reaction time = decision time + 100 ms non-decision time
Recurrent strength $w_+ = 1.68, 1.70, 1.72$	$\alpha = 4.10, 5.50, 8.52\%$; $\beta = 1.31, 1.25, 1.36$
AMPA fraction 0, 25, 50%	$\alpha = 4.79, 6.11, 9.37\%$; $\beta = 1, 1.38, 1.32$ (standard ratio is 15:85)
Bifurcations in μ_0 (at $c' = 0$)	Spontaneous state loses stability at $\mu_0^* \approx 7$ Hz; saddle becomes a symmetric stable state near 47 Hz
Critical coherence	$\approx 70\%$ (saddle annihilates with the less-favored attractor)
Operating regimes in w_+	I: < 1.58; II: 1.58–1.6; III: 1.6–1.86; IV: > 1.86
Saddle time constants	$\tau_{stable} \approx 200$–$300$ ms at small μ_0; $\tau_{unstable}$ of a few seconds in regime II
Numerics	Euler, $dt = 0.1$ ms (checked down to 0.01 ms); 2000 trials per data point; OU time constant 2 ms

2.5 Limitations and open points

- The simulated spontaneous rate (2–3 Hz) is far below that recorded in LIP before stimulus onset (about 15 Hz); the authors attribute this to the presence of the choice targets and note it as incomplete.
- The readout is assumed: a fixed rate threshold, with no biological mechanism for reading it out.
- Trial-history effects are not modeled.
- Regime II (decision without memory) occupies a very narrow range of w_+ and gives low firing rates and overly long reaction times.
- The authors state that the standard parameters may not be optimal for fitting the data.
- The noise level is not explored systematically: the paper only remarks that σ changes how large the coherence must be for perfect accuracy. The reported values also differ across the paper (0.007 nA in Methods, 0.008 nA in the Fig. 3 caption, 0.02 nA in the Appendix model); I use the Appendix value, since that is the model I implement. The noise dimension of the regime map is therefore the clearest gap this project fills.
- The one-dimensional energy landscape (their Fig. 4D) is described as purely illustrative. This supports my decision to use an empirical $-\log P$ landscape and not to claim a true potential.
- Code is available only "on request", so the model has to be re-implemented and checked against the published values above.

2.6 What this means for the project

Seed paper	Decision in this project
Threshold 15 Hz, non-decision time 100 ms	Primary decision criterion; sensitivity analysis over 10–30 Hz and over a fraction of the attractor rate
Appendix parameters, $\sigma = 0.02$ nA, OU time constant 2 ms	Baseline model; sweep σ over about 0.005–0.04 nA
Stimulus $\mu_0 = 30$ Hz	Fixed in all sweeps
2000 trials per point, $dt = 0.1$ ms	1000 trials at baseline, 300 per sweep cell, $dt = 0.5$ ms validated against 0.1 ms (cost control)
Deterministic regime map in (μ_0, w_+)	Regime map in (J_{12}, σ), which adds the noise dimension; the impulsive regime is not in the paper
Saddle time constants, weaker role at large μ_0 and coherence	H1 tested mainly at low coherence; $d \equiv 0$ at $c' = 0$
Critical coherence $\approx 70\%$	Coherences capped at 51.2%, below annihilation; verified with the bifurcation analysis
Illustrative 1D landscape	Empirical $-\log P$ landscape, shown as a static figure
Weibull α, β; errors slower than correct trials	Validation of the baseline implementation before any sweep

2.7 Citation context of the seed paper

Figure 1 shows the citation graph around the seed paper, which appears in the graph under the label "Wong-Lin, 2006" (presumably the first author's later name format). My reading of the graph is that it has three groups of works. The older, lighter nodes on the left are the experimental and behavioral foundations: Roitman and Shadlen (2002), Shadlen (2001), Gold (2001, 2003, 2007), Palmer (2005), Mazurek (2003) and Huk (2005). Around the center are accumulator and drift-diffusion theory: Usher (2001), Smith (2004), Bogacz (2006), Ratcliff (2008) and Brown (2005, 2008). On the right are attractor-network models, including Wang (2002, 2008), Roxin (2008), Furman (2008), Machens (2005) and the later work labelled Wong-Lin (2007) and Wong (2008). The darker, newer nodes at the top (for example Thura 2012 and Eckhoff 2011) are later work that builds on the model, and Brunel (2004, 2014) forms a separate spiking-network cluster at the bottom right. The graph supports the choice of Wang (2002), Bogacz (2006) and Roxin and Ledberg (2008) as the main companion references, and it points to the later papers by the same group as further reading.

3 Model

Each population $i \in \{1, 2\}$ has an NMDA gating variable S_i (the Appendix model of Wong and Wang, 2006, without AMPA at the recurrent synapses):

$$\frac{dS_i}{dt} = -\frac{S_i}{\tau_S} + (1 - S_i)\,\gamma\,H(x_i), \tag{1}$$

$$H(x) = \frac{ax - b}{1 - \exp[-d\,(ax - b)]}, \tag{2}$$

$$x_1 = J_{11}S_1 - J_{12}S_2 + I_0 + J_{A,\text{ext}}\,\mu_0\left(1 + \tfrac{c'}{100}\right) + I_{\text{noise},1}, \tag{3}$$

$$x_2 = J_{22}S_2 - J_{21}S_1 + I_0 + J_{A,\text{ext}}\,\mu_0\left(1 - \tfrac{c'}{100}\right) + I_{\text{noise},2}, \tag{4}$$

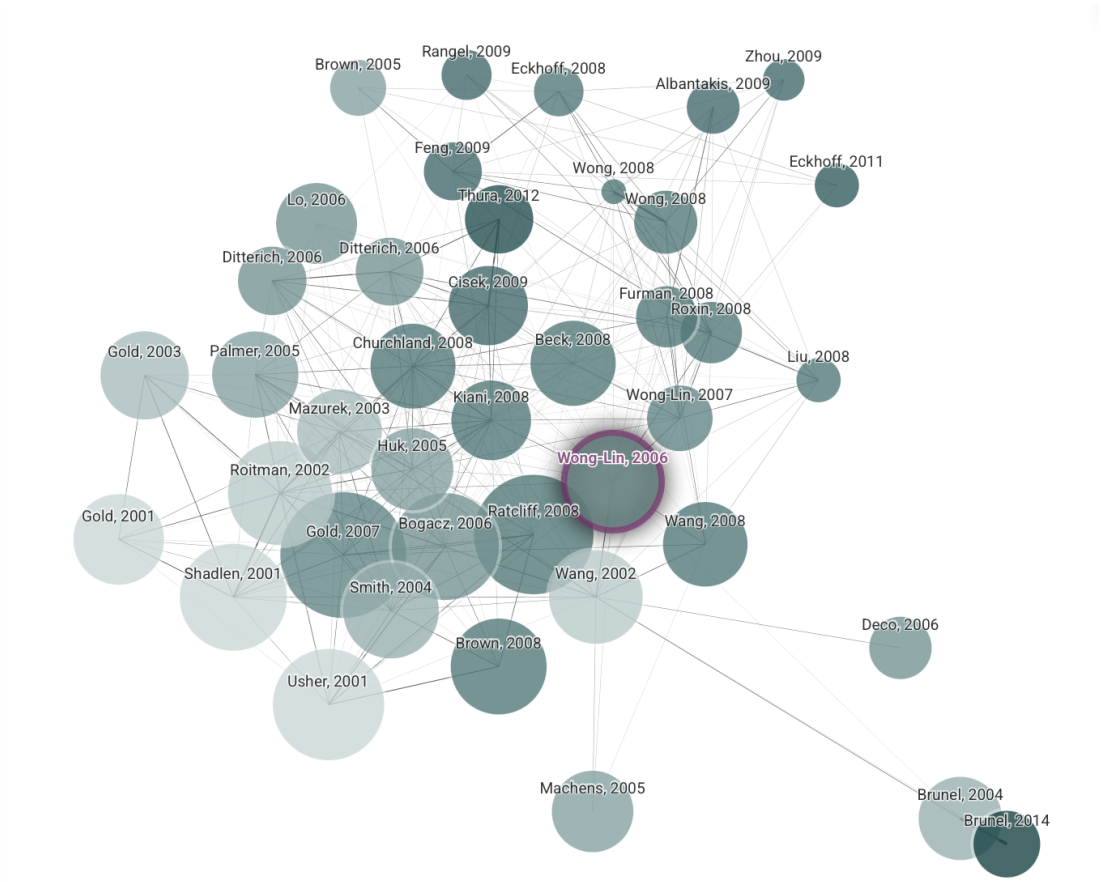


Figure 1: Citation graph around the seed paper (Wong and Wang, 2006), highlighted and labelled "Wong-Lin, 2006". Nodes are papers and edges are citation relations; lighter nodes appear to be older works.

where c' is the signed motion coherence (in percent) toward population 1, $J_{11} = J_{22}$ is the self-excitation, $J_{12} = J_{21}$ the cross-inhibition, I_0 the background input, and $J_{A,\text{ext}}\mu_0$ the external stimulus strength. The firing rate of population i is $r_i = H(x_i)$.

Baseline parameters (Appendix of Wong and Wang, 2006). $\tau_S = 100$ ms, $\gamma = 0.641$, $a = 270$ Hz/nA, $b = 108$ Hz, $d = 0.154$ s, $J_{11} = J_{22} = 0.2609$ nA, $J_{12} = J_{21} = 0.0497$ nA, $J_{A,\text{ext}} = 5.2 \times 10^{-4}$ nA/Hz, $I_0 = 0.3255$ nA, $\mu_0 = 30$ Hz. These values were checked against the Appendix of the paper. Note that the parameter set in the paper's main text ($J_{N,11} = 0.1561$ nA, $J_{N,12} = 0.0264$ nA, $I_0 = 0.2346$ nA) belongs to the version with AMPA at the recurrent synapses and must not be mixed with the values above.

Noise. As in the original model, noise enters as an Ornstein–Uhlenbeck process on the input current:

$$\tau_{\text{AMPA}} \frac{dI_{\text{noise},i}}{dt} = -I_{\text{noise},i} + \sqrt{\tau_{\text{AMPA}}}\,\sigma\,\xi_i(t), \qquad \tau_{\text{AMPA}} = 2 \text{ ms}, \tag{5}$$

with $\xi_i(t)$ unit-variance Gaussian white noise and baseline $\sigma = 0.02$ nA (Appendix value). This also keeps S_i within $[0, 1]$ naturally.

4 Methods

Deterministic analysis. Nullclines, vector fields, fixed points (root finding with continuation), Jacobian eigenvalues, and the separatrix (the stable manifold of the saddle, obtained by backward integration). Bifurcation diagrams are computed over J_{12} and coherence; this step is deterministic and computationally cheap.

Stochastic simulation. Euler–Maruyama integration (Higham, 2001) with $dt = 0.5$ ms, validated against $dt = 0.1$ ms (the step used in the seed paper) on a subset. Trials begin at the spontaneous state with a 500 ms pre-stimulus period, followed by stimulus onset, with a maximum trial duration of 2 s. The Monte Carlo code is vectorized across trials and compiled with Numba, using fixed random seeds.

Decision criterion. The primary criterion follows the seed paper: a decision is made when the winning population's rate first reaches $\theta = 15$ Hz (about $S \approx 0.49$), the decision time T_D is measured from stimulus onset, and reaction time is $T_D + 100$ ms. Because the choice-attractor rates are not given explicitly for the Appendix parameter set, I first check that θ lies well below them. Sensitivity of all results is shown over $\theta \in [10, 30]$ Hz and for a criterion defined as a fixed fraction (e.g. 70–80%) of the attractor rate. Trials are labeled correct, error, or no-decision (timeout).

Experimental design.

- Coherence levels: 0, 3.2, 6.4, 12.8, 25.6, 51.2% (standard in Roitman and Shadlen, 2002); all below the critical coherence of about 70%.
- Baseline experiments: 1000 trials per coherence level.
- Parameter sweeps: a 15×15 grid over (J_{12}, σ) with 300 trials per cell at two coherences (0% and 12.8%). This is on the order of 10^8 time steps, feasible in minutes to an hour with Numba.

Baseline validation. Before any sweep, the implementation is checked against the seed paper: a Weibull fit $P_c(c') = 1 - 0.5\exp[-(c'/\alpha)^\beta]$ of the psychometric function (reference $\alpha = 7.2\%$, $\beta = 1.25$), errors slower than correct trials, and a decision-time decrease with coherence.

Regime definitions (operational).

- *Normal:* accuracy rises and RT falls with coherence.
- *Impulsive:* short RT with accuracy near chance even at high coherence.
- *Indecisive:* no-decision rate above a preset criterion.

The indecisive regime corresponds to the paper's regime I (no competition reachable) and the paper's regime IV (no competition because a symmetric stable state always coexists) provides a deterministic counterpart at strong self-excitation. The impulsive regime is not described in the paper.

Quasi-potential. Because the system is non-gradient, no true potential is defined. Instead, $-\log P$ estimated from long simulations (a 2D density plus a 1D cut along $S_1 - S_2$) is shown as an empirical landscape in a static figure.

5 Work Plan (12 Weeks)

Weeks	Work	Deliverable
1–3	Deterministic model, nullclines, fixed points, saddle and separatrix, unit tests	Phase portrait with separatrix
4–6	OU-noise simulations, baseline validation against the seed paper, psychometric and chronometric curves, error vs. correct RT; start Methods and Results drafting	Validated baseline behavior
7–9	Bifurcation analysis over J_{12}; (J_{12}, σ) sweeps; regime map; test H1 and H2	Phase diagram heatmap and regression results
10–12	Empirical landscape figure, threshold and dt sensitivity, final report, repository cleanup	Report (5–10 pages) and reproducible repository

The report is written continuously from Week 4 onward. Weeks 10–12 are for polishing, not first drafts.

6 Expected Results and Validation

Baseline validation should show a monotonic increase of accuracy with coherence, decreasing RT with coherence, errors slower than correct trials, and trajectories passing near the saddle's unstable manifold. Validation is against the qualitative and quantitative results of Wong and Wang (2006) (Table in Section 2) and the neural and behavioral signatures in Roitman and Shadlen (2002).

For the parameter sweeps, I expect a normal regime in an intermediate range, flanked by:

- **Indecisive behavior** when cross-inhibition is too strong or recurrent excitation too weak for the choice attractors to exist (or to be reached in time).
- **Impulsive behavior** when noise is high relative to the barrier around the spontaneous state, or when the spontaneous state loses stability. High noise, not low noise, drives fast, evidence-independent choices; very low noise instead lengthens RTs and can cause timeouts.

This two-sided pattern is consistent in spirit with the inverted-U dependence of performance on E/I balance reported in the spiking model by Lam et al. (2022), and I will compare against it. Regions with unexpected structure (e.g. multiple stable states) will be reported as findings and analyzed with the bifurcation tools.

7 Scope Boundaries and Optional Extensions

Out of scope unless the core plan is complete by Week 9: a drift-diffusion fit to the simulated RT distributions, a perseveration protocol (which requires evidence that changes mid-trial), an animated landscape, and a cross-check against the spiking network.

8 Deliverables

1. A documented Python package (deterministic system, stochastic integrator, phase-plane analysis, experiment runners, plotting).
2. Three core figures: phase portrait with separatrix; psychometric and chronometric curves; (J_{12}, σ) regime map with bifurcation curves overlaid. Plus one supplementary empirical-landscape figure.

3. A 5–10 page LaTeX report.

4. A public repository with fixed seeds and a one-command script reproducing every figure.

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